

FitBuddy AI — Scalability & Future Plan

Phase 8 • Project Demonstration

Current architecture foundation

- FastAPI + SQLite + HTML/CSS/JavaScript provides a simple local-first base.
- The separation of routes, services, schemas, models, database handling, and frontend assets provides clear areas for future extension.

Future directions

- Expand plan-adjustment rules and feedback categories.
- Enhance coach summaries and operational reporting.
- Add broader automated test coverage and record measured performance results.
- Reconcile the Gemini dependency before relying on live integration.
- Move from local SQLite to a larger deployment database if future scale requires it, while retaining the service/API boundaries.

Scaling considerations

- Keep configuration externalized through environment variables.
- Keep API contracts stable as frontend and coach features grow.
- Separate generation logic from request handling so generation strategies can evolve.